Bathymetry + ENC Drop

Replaces yesterday's hand-eyeballed shoreline mask with proper depth-and-hazard-based navigability. This is the foundation for safe routing everywhere, not just Lake Michigan western shore.

Files

File	Status	Purpose
`backend/app/services/boats.py`	NEW	Per-class spec table — polar + draft + LOA + displacement
`backend/app/services/bathymetry/__init__.py`	NEW	DepthGrid + GCS loader with bilinear sampling
`backend/app/services/charts/__init__.py`	NEW	HazardIndex with shapely STRtree
`backend/app/services/routing/navigability.py`	NEW	Combines depth + hazards into the engine's predicate
`backend/app/services/routing/__init__.py`	🔄	Exports the predicate builder
`backend/app/routers/routing.py`	🔄	Uses boat spec lookup + predicate; returns 503 on missing bathy
`backend/workers/bathymetry_ingest.py`	NEW	NCEI Lake MI + 9 CRM volumes → packed npz on GCS
`backend/workers/enc_ingest.py`	NEW	ENC Direct REST → merged GeoJSON on GCS
`backend/tests/test_bathymetry.py`	NEW	7 tests, all green
`backend/tests/test_navigability.py`	NEW	5 tests, all green
`backend/requirements.txt`	🔄	Adds `netCDF4`, `shapely`
`docs/2026-05-05-bathymetry-rollout.md`	NEW	Runbook for the workers

The previous `coastlines.py` should be deleted from your repo — `navigability.py` replaces it entirely.

What changed in the API surface

`POST /api/routing/compute` now accepts an optional `safety_factor` (1.0–3.0, default 1.5):

```
{ "race_id": "...", "safety_factor": 1.5 }
```

The response `meta` block now includes `boat_class`, `draft_m`, and `min_depth_m` so the UI can show "routed for a Beneteau 36.7 with 6'11" draft, avoiding water shallower than 10.4 ft."

If bathymetry hasn't been ingested for the route's region, the endpoint returns **503** with a clear message ("Run bathymetry_ingest for this region before computing routes"). This is intentional — failing-open silently routes through land.

Cache invalidation

Engine version bumped to `v3-bathy`. Cache key is now `route:v3-bathy:{race_id}:{ref_time}:{safety_factor}` so any route cached under the previous shoreline-mask version is invisible to the new code, automatically.

Critical path before Saturday

1. **Push backend.** Frontend doesn't change — keeps using the same endpoint with the same payload shape.
2. **Run `bathymetry_ingest.py --region conus --source ncei_great_lakes`.** This is required. Without it, every route compute returns 503.
3. **Run `enc_ingest.py --region conus`.** Optional but recommended; adds breakwaters, restricted areas, etc. that bathymetry alone misses.
4. **Test recompute.** Click Compute Route on the Waukegan→Chicago race; expect a magenta line that hugs the shore at Wilmette without crossing it.

Detailed steps in `docs/2026-05-05-bathymetry-rollout.md`.

Known gaps (for after Saturday)

- **Per-volume CRM ingest overwrites the same key.** Running `ncei_crm_vol1` would clobber the Lake Michigan grid. Need to either split into per-coast region keys or merge volumes before upload. Documented in the runbook.
- **No water-level / tide correction.** NCEI grids reference LWD/MLLW. Lake Michigan is typically 0.3–0.8 m above LWD, ocean tides ±2 m. For Saturday's open-lake passage the

route is in 50+ m of water and this doesn't matter. For harbor approaches it does.

- **ENC Direct layer IDs assumed stable.** If NOAA renumbers the service, layers return 0 features (worker logs a warning). Update the `HAZARD_LAYERS` list in `enc_ingest.py`.
- **No ENC point/line features.** Buoys, cables, individual rocks marked as points are skipped at v1. Polygons (LNDARE etc.) cover the important cases.
- **Coastlines.py is now unused.** Delete it from the repo when cherry-picking these changes; it's referenced by nothing.